

1. Prepare each file and add explicit links

VCF inputs (four scenarios)

1: Five gene-span slices:

NB72462M,
NG131FQA1I, 0GOOR_HG002,
HG004_GRCh38, HG005_GRCh38
(all .vcf.gz)

2: 104 single-sample cohort VCFs

3: Complete HG005_GRCh38 VCF

4: Complete NB72462M VCF

+ ClinVar; + frequencies (Scenario 2)
Then normalize (GRCh38).

VCF-RDfizer conversion

VCF → RDf graph per file.

Link plug-in: same VCF

Rules: linker.ttl
Pinned GRCh38 references
SPDI from position + alleles
Gene interval overlap
→ separate .links.nt.

RDf graphs + their link triples

Shared variant identity:

participant call → SPDI ← ClinVar call
frequency-panel call → the same SPDI
Edges: *vcfl:sameVariantAs*

Gene links for policy selection:

participant call → Ensembl gene identifier
Edge: *vcfl:overlapsGene*
Shared identifiers let queries join the files.

2. Build a validated release view

Where the policy is defined

Case JSON + gene list
→ generate.py → policy.ttl
ODRL file consents and withdrawal;
28 cancer genes: clinical care only.
Scenario 4: + panel, region, variant.
Purposes use DUO terms.

Policy plug-in: evaluate

Select file records and gene overlaps.
Match requester and purpose.
A prohibition overrides permission.
No permission → withhold.
Write permitted data and links.

Request context

--assignee: requester identifier
--purpose: DUO purpose term
CC, DS, and GRU requesters,
plus OP in Scenario 4.

participant RDf + links

release view

Policy plug-in: check

Compare with a VCF-derived reference.
Check ownership and dangling links.
Pass → validated view.nt.gz
One release view per requester.

shared annotation RDf + links

3. Query each release view

SPARQL query enters here

carriers.rq: 81 ACMG genes;
ALT carrier; PASS/missing FILTER;
ClinVar review; join calls by SPDI.
rare.rq: add frequency < 1%.

QLever indexes and queries

Validated participant release view
+ ClinVar RDf and links
+ frequency RDf and links (Scenario 2).
Run the same query for each requester.

Returned information

File, sample, gene, chromosome,
position, REF/ALT, SPDI,
ClinVar classification, genotype;
and frequency for rare.rq.